

Proof of $BB_{SK}(7) = 41$

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This note proves that the maximum normal form size produced by any terminating SK -combinator expression of length 7 is 41. The argument combines exhaustive enumeration of all length-7 terms with a structural non-termination proof for the only remaining unresolved candidates.

1 Exhaustive Space Enumeration and Filtering

To establish the exact value of $BB_{SK}(7) = 41$, we first execute a two-phase filtering process: exhaustive state-space enumeration followed by bounded normal-order execution.

1.1 Combinator State Space Enumeration

A combinator term of length n is defined as a binary expression tree containing exactly n leaf nodes, where each leaf node is an element of the alphabet $\Sigma = \{S, K\}$. The structure of the expression is determined by the arrangement of application nodes.

The total number of unlabeled strictly binary trees with n leaves is given by the $(n - 1)$ -th Catalan number, C_{n-1} . For each such tree structure, every leaf node has two independent choices (S or K). Therefore, the total size of the search space for terms of length n is

$$\text{TotalTerms}(n) = 2^n C_{n-1}.$$

For $n = 7$, the sixth Catalan number is 132. The total number of valid combinator terms of length 7 is therefore

$$2^7 \cdot 132 = 16896.$$

1.2 Bounded Normal-Order Filtering

A decider program evaluates all 16896 terms using leftmost-outermost (normal-order) reduction for at most 15 steps. A reference implementation of the enumeration and filtering strategy, as well as the results, are available in the accom-

panying GitHub repository: <https://github.com/b00ne/bb-sk-7>. The terms are partitioned into two categories:

- Halting terms (≤ 15 steps): terms that successfully reduce to a normal form (a state containing no further reducible expressions, or redexes) within 15 steps. The maximal combinator-count normal form observed among all terminating terms in this space is 41 combinators, achieved by the normal form of the term $SSS(S(SS))S$.
- Unresolved terms: terms that have not reduced to a normal form after 15 reduction steps.

Out of the 16896 total candidates, exactly two remain unresolved after 15 steps of reduction:

$$\begin{aligned} T_1 &= S(SS)SSSS, \\ T_2 &= SSS(SS)SS. \end{aligned}$$

1.3 Equivalence via a Single Reduction Step

Notation: normal-order reduction. Throughout this proof, the reduction arrow $\xrightarrow{n}$ and step count n refer strictly to standard normal-order (leftmost-outermost) reduction, unless otherwise specified.

One-step equivalence. It is immediate that

$$T_1 \xrightarrow{1} SSS(SS)SS = T_2.$$

Because T_1 reduces to T_2 in exactly one normal-order step, their termination behaviours coincide:

$$T_1 \text{ is divergent} \iff T_2 \text{ is divergent}.$$

Consequently, proving the termination behaviour of $T_1 = S(SS)SSSS$ simultaneously settles the outcome for both unresolved candidates.

2 Proof of Non-Termination for $S(SS)SSSS$

2.1 Proof Strategy

To prove divergence of $T_1 = S(SS)SSSS$, we construct an infinite family of expressions M_n such that each instance reduces into a larger structure containing the next instance. By establishing that the initial term $S(SS)SSSS$ can be reduced to an expression of the form FM_0 (where F and M_0 are explicitly defined combinator terms), and that each FM_n reduces to a subsequent application FM_{n+1} , we demonstrate that $S(SS)SSSS$ itself admits an infinite reduction path.

2.2 Preliminaries: contexts and reduction

Contexts. A *context* $\mathcal{C}[\cdot]$ is a combinator expression with a single distinguished hole. We write $\mathcal{C}[t]$ for the result of replacing the hole by t .

Reduction relation (definition). Let $\mathcal{R}$ be the set of primitive rewrite rules; in our setting $\mathcal{R}$ contains the single S -rule

$$S\ x\ y\ z \rightarrow x\ z\ (y\ z).$$

We define the one-step reduction relation $\rightarrow$ by two clauses:

1. **Primitive steps.** Every primitive rule is a one-step reduction: if $\ell \rightarrow r'$ is a rule in $\mathcal{R}$ then $\ell \rightarrow r'$.
2. **Contextual compatibility.** Reduction is compatible with application of contexts: for all terms t, t' and every context $\mathcal{C}[\cdot]$, if $t \rightarrow t'$ then $\mathcal{C}[t] \rightarrow \mathcal{C}[t']$.

Formally, $\rightarrow$ is the *smallest* relation satisfying (1) and (2). Intuitively, a single reduction step may be either an application of a primitive rule at the top level or the same primitive step performed at any position inside a larger term.

Contextual closure (many-step). Write $\rightarrow^*$ for the reflexive transitive closure of $\rightarrow$ (zero or more $\rightarrow$ steps). Because $\rightarrow$ is contextually compatible, reduction sequences are preserved when placed inside any fixed surrounding expression. Concretely, if $A \rightarrow^* B$ then for every context $\mathcal{C}[\cdot]$ we have

$$\mathcal{C}[A] \rightarrow^* \mathcal{C}[B].$$

This contextual closure is standard; see [1, 2].

Lemma 1 (Persistence of divergence in pure S). *Let E be a pure S -combinator expression (no K or other erasing combinators). If E contains a subterm T that has no normal form (i.e., T is divergent), then E has no normal form.*

Proof. Write $E = \mathcal{C}[T]$ for a context $\mathcal{C}[\cdot]$. By assumption T admits an infinite reduction sequence

$$T \rightarrow T_1 \rightarrow T_2 \rightarrow \dots$$

Since $\rightarrow$ is closed under contexts, each step $T_i \rightarrow T_{i+1}$ yields a step $\mathcal{C}[T_i] \rightarrow \mathcal{C}[T_{i+1}]$. The pure S rule

$$S\ x\ y\ z \rightarrow x\ z\ (y\ z)$$

is non-erasing (every argument occurrence appears on the right), so no finite sequence of outer reductions can remove the occurrence of the divergent subterm. Hence the steps form an infinite reduction sequence for $E = \mathcal{C}[T]$, and E has no normal form. $\square$

2.3 Preliminary Definitions

Let x be a fixed dummy variable. Define

$$\begin{aligned} F &= S(S(SS)S)S, \\ D &= S(S(S(S(SS)F))), \\ M_0 &= Dx, \\ M_n &= SM_{n-1}(x(SM_{n-1})) \quad \text{for } n \geq 1. \end{aligned}$$

2.4 Inductive Cycle

We establish that the family of expressions M_n forms an infinite reduction cycle by breaking the transformation down into three phases.

Notation: evaluation contexts. Throughout this section, each $\mathcal{C}_i[\cdot]$ denotes a fixed evaluation context (a combinator expression with a single distinguished hole, as in Section 2.2). The exact shapes of these contexts are unimportant; the argument applies regardless of their internal structure.

Phase A: Expansion of FM_n . For any n , the application of FM_n reduces to reveal a self-application pattern:

$$FM_n \xrightarrow{*} \mathcal{C}_0[M_n(SM_n)].$$

By expanding the definition of F and applying it to an arbitrary term t , a direct three-step normal-order reduction gives

$$Ft \xrightarrow{3} \mathcal{C}_0[t(St)].$$

(See the Trace Appendix for the three-step trace of Ft .)

Substituting M_n for t produces the claimed context.

Phase B: Reduction to the base term. Any expression M_n applied to SM_n reduces through successive applications of M_n to the base term:

$$M_n(SM_n) \xrightarrow{*} \mathcal{C}_1[M_0(SM_n)].$$

By definition,

$$M_n = SM_{n-1}(x(SM_{n-1})).$$

Applying this definition to an arbitrary term t and performing one reduction yields

$$M_n t \xrightarrow{1} M_{n-1} t(x(SM_{n-1})t).$$

Every application of $M_n t$ thus structurally reduces to an application of $M_{n-1} t$, and induction on n shows the expression expands until it reaches the base term M_0 . Substituting SM_n for t gives the claimed target context.

Phase C: Index advancement. The base term M_0 increments the family index when it is applied to SM_n :

$$M_0(SM_n) \xrightarrow{*} C_2[FM_{n+1}].$$

Using $M_0t = Dxt$, after four reduction steps the base application produces the term $F(t(xt))$:

$$Dxt \xrightarrow{4} C_2[F(t(xt))].$$

(See the Trace Appendix for the four-step trace of Dxt .)

Substituting SM_n for t gives

$$M_0(SM_n) \xrightarrow{*} C_2[F(SM_n(x(SM_n)))].$$

By the definition of the inductive family, the subterm $SM_n(x(SM_n))$ is exactly M_{n+1} , so the reduction simplifies to

$$M_0(SM_n) \xrightarrow{4} C_2[FM_{n+1}].$$

Lemma 2. *For every $n \geq 0$, the term FM_n reduces to a term of the form FM_{n+1} , up to a surrounding context; that is, there exists a context $\mathcal{C}$ such that*

$$FM_n \rightarrow^* \mathcal{C}[FM_{n+1}].$$

Proof. Compose the three phases: Phase A expands FM_n to $C_0[M_n(SM_n)]$, Phase B expands $M_n(SM_n)$ to $C_1[M_0(SM_n)]$, and Phase C advances the index to $C_2[FM_{n+1}]$. Since each phase is independent of n , their composition yields the required steps from FM_n to FM_{n+1} for every n . $\square$

By repeated application of Lemma 2, the family produces an infinite sequence of index increments

$$C_a[FM_0] \xrightarrow{*} C_b[FM_1] \xrightarrow{*} C_c[FM_2] \xrightarrow{*} \dots,$$

showing that the subterm FM_0 admits an infinite reduction path.

2.5 Initialization and Main Theorem

Theorem 1. *The pure S -combinator term $S(SS)SSSS$ is divergent (has no normal form).*

Proof. Begin by executing the first nine normal-order reduction steps of the initial term:

$$S(SS)SSSS \xrightarrow{9} S(S(SS)F)(F(S(SS)F)).$$

(See the Trace Appendix for the nine-step trace of $S(SS)SSSS$.)

Write the left-hand subterm as $\mathcal{L} = S(S(SS)F)$ and the right-hand subterm as $\mathcal{R} = F(S(SS)F)$. To form a valid redex, S must be supplied

three arguments. In $\mathcal{LR}$ the outermost head is the S of $\mathcal{L}$: it already has the internal argument $S(SS)F$, and the top-level application supplies $\mathcal{R}$, so it has exactly two arguments and thus does not form a valid redex; likewise the internal $S(SS)F$ has only two outermost arguments and is inert.

Because the left branch $\mathcal{L}$ is in normal form and cannot be reduced, normal-order reduction is forced to evaluate only the right-hand subterm. The reduction of $\mathcal{R}$ can be described as a sequence of localized normal-order reductions inside progressively nested contexts $\mathcal{K}_i[\cdot]$, as follows:

$$F(S(SS)F) \xrightarrow{3} \mathcal{K}_0[S(SS)F(S(S(SS)F))],$$

$$S(SS)F(S(S(SS)F)) \xrightarrow{6} \mathcal{K}_1[S(S(SS)F)(S(S(S(SS)F)))(S(S(S(SS)F)))],$$

$$S(S(SS)F)(S(S(S(SS)F)))(S(S(S(SS)F))) \xrightarrow{8} \mathcal{K}_2[S(S(S(SS)F))DD],$$

$$S(S(S(SS)F))DD \xrightarrow{3} \mathcal{K}_3[F(DD)].$$

(See the Trace Appendix for the detailed piecewise reductions used above.)

In each case the selected subterm is evaluated in isolation, while the surrounding structure is preserved by the corresponding context $\mathcal{K}_i[\cdot]$. The subterm DD represents the base term $M_0 = Dx$ with the dummy variable x instantiated as D . Since the behavior of the inductive cycle does not depend on the specific choice of x , we may write this term as M_0 , yielding

$$F(S(SS)F) \xrightarrow{*} \mathcal{K}_3[FM_0].$$

We have proven inductively via Lemma 2 that the subterm FM_0 possesses an infinite reduction path. By Lemma 1, because the pure S -calculus cannot erase subexpressions, the existence of this infinite reduction path within $\mathcal{K}_3[FM_0]$ guarantees that the parent expression $\mathcal{R}$ has no normal form. Consequently, the initial term $S(SS)SSSS$ is divergent. $\square$

3 Conclusion

By coupling the exhaustive filtering with the structural induction proof showing that T_1 expands into an infinitely growing cycle

$$\mathcal{C}_a[FM_0] \xrightarrow{*} \mathcal{C}_b[FM_1] \xrightarrow{*} \mathcal{C}_c[FM_2] \xrightarrow{*} \dots,$$

we prove that the only two unresolved terms in the length-7 term space are divergent. Because all other length-7 candidates halt within 15 steps, the term $SSS(S(SS))S$, which reduces to a normal form of length 41, attains the maximum normal form size for $n = 7$. Hence

$$BB_{SK}(7) = 41,$$

References

- [1] J. R. Hindley and J. P. Seldin, *Introduction to Combinators and λ -Calculus*, Cambridge University Press, 1986.
- [2] Terese (ed.), *Term Rewriting Systems*, Cambridge Tracts in Theoretical Computer Science, Cambridge University Press, 2003.

A Reduction Trace Appendix

This appendix provides step-by-step normal-order reduction traces for the key terms cited in the proof. Each reduction is presented as an ordered sequence of terms showing the successive states of the expression.

Trace 1: Ft (expansion of F)

- 1 Ft
- 2 $S(SS)St(St)$
- 3 $SSt(St)(St)$
- 4 $S(St)(t(St))(St)$

Legend: The reduction of Ft shows how the definition $F = S(S(SS)S)S$ expands under application to an arbitrary term t , yielding the self-application pattern $t(St)$.

Trace 2: Dxt (expansion of D)

- 1 Dxt
- 2 $S(S(S(SS)F))t(xt)$
- 3 $S(S(SS)F)(xt)(t(xt))$
- 4 $S(SS)F(t(xt))(xt(t(xt)))$
- 5 $SS(t(xt))(F(t(xt)))(xt(t(xt)))$

Legend: The reduction of Dxt shows the expansion of the base term constructor, ultimately producing the form $F(t(xt))$ required for the inductive cycle.

Trace 3: $S(SS)SSSS$ (initialization)

Normal-order reduction of the initial term $S(SS)SSSS$ for 9 steps:

- 1 $S(SS)SSSS$
- 2 $SSS(SS)SS$
- 3 $S(SS)(S(SS))SS$
- 4 $SSS(S(SS)S)S$
- 5 $S(S(SS)S)(S(S(SS)S))S$
- 6 $S(SS)SSF$
- 7 $SSS(SS)F$

8 $S(SS)(S(SS))F$
9 $SSF(S(SS)F)$
10 $S(S(SS)F)(F(S(SS)F))$

Legend: The reduction of $S(SS)SSSS$ shows how the initial term expands in 9 steps to reach the form $S(S(SS)F)(F(S(SS)F))$, which splits into the left-normal-form component and the right-hand term that expands to $\mathcal{C}_a[F(DD)]$.

Trace 4: $F(S(SS)F)$ (piecewise)

The reduction of $F(S(SS)F)$ is presented as a sequence of smaller staged evaluations. Each stage focuses on a selected subterm while preserving the surrounding structure in a context $\mathcal{K}_i[\cdot]$.

1 $F(S(SS)F)$
2 $S(SS)S(S(SS)F)(S(S(SS)F))$
3 $SS(S(SS)F)(S(S(SS)F))(S(S(SS)F))$
4 $S(S(S(SS)F))(S(SS)F(S(S(SS)F)))(S(S(SS)F))$
5 $S(S(SS)F)(S(S(SS)F))(S(SS)F(S(S(SS)F))$
 $(S(S(SS)F)))$

which yields $\mathcal{K}_0[(S(SS)F(S(S(SS)F)))]$.

1 $S(SS)F(S(S(SS)F))$
2 $SS(S(S(SS)F))(F(S(S(SS)F)))$
3 $S(F(S(S(SS)F)))(S(S(SS)F)(F(S(S(SS)F))))$
4 $S(S(SS)S(S(S(SS)F))(S(S(S(SS)F))))$
 $(S(S(SS)F)(F(S(S(SS)F))))$
5 $S(SS(S(S(SS)F))(S(S(S(SS)F)))(S(S(S(SS)F))))$
 $(S(S(SS)F)(F(S(S(SS)F))))$
6 $S(S(S(S(SS)F))(S(S(S(SS)F)))(S(S(SS)F)$
 $(S(S(S(SS)F)))(S(S(S(SS)F))))(S(S(SS)F)(F(S(S(SS)F))))$

which yields $\mathcal{K}_1[(S(S(SS)F)(S(S(S(SS)F)))(S(S(S(SS)F))))]$.

1 $S(S(SS)F)(S(S(S(SS)F)))(S(S(S(SS)F)))$
2 $S(SS)F(S(S(S(SS)F)))(S(S(S(SS)F))(S(S(S(SS)F))))$
3 $SS(S(S(S(SS)F)))(F(S(S(S(SS)F)))(S(S(S(SS)F))(S(S(S(SS)F))))$
4 $S(F(S(S(S(SS)F)))(S(S(S(SS)F))(F(S(S(S(SS)F))))$
 $(S(S(S(SS)F))(S(S(S(SS)F))))$
5 $F(S(S(S(SS)F)))(S(S(S(SS)F))(S(S(S(SS)F))))$
 $(S(S(S(SS)F))(F(S(S(S(SS)F)))(S(S(S(SS)F))(S(S(S(SS)F))))$
6 $S(SS)S(S(S(S(SS)F))D(S(S(S(SS)F))(S(S(S(SS)F))))$
 $(S(S(S(SS)F))(F(S(S(S(SS)F)))(S(S(S(SS)F))(S(S(S(SS)F))))$
7 $SS(S(S(S(SS)F))DD(S(S(S(SS)F))(S(S(S(SS)F))))$
 $(S(S(S(SS)F))(F(S(S(S(SS)F)))(S(S(S(SS)F))(S(S(S(SS)F))))$
8 $DD(S(S(S(SS)F))DD)(S(S(S(SS)F))(S(S(S(SS)F))))$
 $(S(S(S(SS)F))(F(S(S(S(SS)F)))(S(S(S(SS)F))(S(S(S(SS)F))))$

which yields $\mathcal{K}_2[(S(S(S(SS)F))DD)]$.

- 1 $S(S(S(SS)F))DD$
- 2 $S(S(SS)F)D(DD)$
- 3 $S(SS)F(DD)(D(DD))$
- 4 $SS(DD)(F(DD))(D(DD))$

which yields $\mathcal{K}_3[(F(DD))]$.

Legend: The reduction shown above presents each focused stage of the evaluation. The surrounding expressions are abstracted by the contexts $\mathcal{K}_0, \mathcal{K}_1, \mathcal{K}_2, \mathcal{K}_3$.